

Algebraic Traps on the Power Curve

New Progress Toward the Alaoglu–Erdős Conjecture

Deterministic Coppersmith–Chebyshev Certificates,
Sparse Total Positivity, and the High-Corank Frontier

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Bottom line. This report does not claim a proof of the Alaoglu–Erdős conjecture. It develops a new and fully quantitative route prompted by the June 2026 work of Brisebarre and Hanrot on Coppersmith-style detection of integer points near transcendental curves. A hypothetical counterexample supplies, in every logarithmic unit block, an initial segment of an irrational rotation orbit on the fixed arc $y = x^{\log_2 3}$, sampled on the rational grid with denominators 2^H and 3^H . On this particular curve, the usual “two auxiliary polynomials must be coprime” heuristic can be removed: restrictions of sparse bivariate polynomials are exponential sums with distinct real frequencies, hence form a strictly totally positive Chebyshev system. If r linearly independent auxiliary polynomials with a common q -monomial support are uniformly smaller than one, then the block contains at most $q - r$ exact integer points.

A one-dimensional Chebyshev lattice, optimized over the q lowest frequencies $i \log 2 + j \log 3$, then gives the unconditional block estimate

$$\#\{x \in [H, H+1) : 2^x, 3^x \in \mathbb{Z}\} \leq \left(\frac{32 \log 2 \log 3}{9} + o(1) \right) \frac{H^2}{(\log H)^2}.$$

The logarithmic-square saving over a quadratic block bound is genuine new progress, but it still does not contradict the linear block population forced by a counterexample. Optimizing macroscopic windows makes the residual barrier exact: with the current kernel-level lower bound $\beta \geq 14$ for a least counterexample, a closing trap would need to certify more than 98.59% of the entire coefficient space as uniformly short. The remaining problem is thus not “find one clever auxiliary polynomial” but “prove a near-full-rank collapse of a structured diagonal-flow lattice.” The report formulates this as a parametric-geometry-of-numbers problem, proves a dilation-rigidity theorem that constrains multiscale reuse, and gives a proof-producing finite-certificate design suitable for exact arithmetic and Lean replay.

All new assertions are explicitly tagged. No numerical reconnaissance is promoted to a theorem.

Abstract

Let

$$I(H) := \#\{x \in [H, H+1) : 2^x \in \mathbb{Z}, 3^x \in \mathbb{Z}\}.$$

The attached unified report already develops the standard transcendence reductions, the exact conditional solution monoid, determinant and p -adic methods, Kummer and cyclotomic reformulations, compact-orbit dynamics, rational-point counting, and a large collection of quantified no-go theorems. The present document deliberately imports rather than repeats that material.

Its first new observation is algorithmic: after rescaling the H th block by $(2^H, 3^H)$, all exact points lie on one fixed analytic arc and fit the rational-grid problem studied by Brisebarre–Hanrot. Its second observation is structural: for any finite monomial support $E \subset \mathbb{N}_0^2$, the functions $e^{(i \log 2 + j \log 3)t}$, $(i, j) \in E$, form an extended complete Chebyshev system. This turns short auxiliary vectors directly into deterministic point-counting certificates and replaces coprimality by linear algebra. Its third observation is asymptotic: a one-variable Chebyshev expansion along the exact curve, combined with Cauchy–Binet and Minkowski’s second theorem, produces short integer polynomials. The asymptotically optimal support is the weighted simplex of the lowest frequencies, not the usual total-degree triangle. This yields

$$\limsup_{H \rightarrow \infty} I(H) \frac{(\log H)^2}{H^2} \leq \frac{32 \log 2 \log 3}{9} = 2.7075555926 \dots$$

For a window $[H, (1+r)H]$, the same calculation has a sharp variational constant. Writing t_0 for the positive root of $t \tanh t = 2$, the best coefficient-space-to-counterexample population ratio is

$$\kappa_* = \frac{64 \log 2 \log 3}{9} \frac{\cosh t_0}{t_0^2} = 5.0872665807 \dots$$

Thus a least counterexample $\beta \geq 14$ would require certified short-vector rank fraction strictly exceeding $1 - 1/(14\kappa_*) = 0.9859593 \dots$. This high-corank requirement is the principal new frontier. We connect it to Schmidt–Summerer–Roy parametric geometry of numbers, prove exact dilation rigidity for sparse trap spaces, analyze why naive Hermite multiplicity is not free, and specify a deterministic certificate format whose verification uses only integer linear algebra and interval bounds.

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1 Executive summary and status ledger

1.1 What is new in this report

The mathematical contributions are the following.

1. **Exact curve-grid interface.** Each logarithmic block is identified with exact integer points on a fixed power arc sampled at the rational grid $(2^{-H}\mathbb{Z}) \times (3^{-H}\mathbb{Z})$. This is the precise interface with the 2026 Coppersmith–Chebyshev method.
2. **Sparse Chebyshev rigidity.** A q -monomial polynomial restricts to an exponential sum with q distinct real frequencies and therefore has at most $q - 1$ zeros on the positive power curve, counted with multiplicity.
3. **Rank-codimension certificates.** If r independent integer polynomials with one q -monomial support are uniformly smaller than one on an interval of the curve, then that interval contains at most $q - r$ exact integer points. No resultant and no coprimality assumption is required.
4. **An optimized unconditional block theorem.** A one-dimensional Chebyshev lattice and a weighted-frequency simplex give

$$I(H) \leq \left(\frac{32 \log 2 \log 3}{9} + o(1) \right) \frac{H^2}{(\log H)^2}.$$

This is a paper proof, not a numerical heuristic.

5. **A sharp high-corank threshold.** The best macroscopic-window constants are solved explicitly. The current finite exclusion $\beta \geq 14$ forces any closing certificate of this type to find at least 98.59% of all coefficient directions below the shortness threshold.
6. **A multiscale rigidity package.** Trap spaces cannot recur exactly under the $(X, Y) \mapsto (2X, 3Y)$ dilation unless they are monomial. This identifies the precise amount of approximate covariance that a parametric-geometry argument would need.
7. **A deterministic certificate design.** The report gives an exact output format for interval-arithmetic Chebyshev bounds, integer coefficient vectors, and rank checks. Such a certificate proves a finite block bound without trusting the LLL implementation that found it.

1.2 What is not claimed

No theorem below proves that a counterexample cannot exist. In particular:

- the new $H^2/(\log H)^2$ block bound is still much larger than the conditional $H/\beta + O(1)$ population;
- no near-full-rank successive-minimum theorem is proved;
- the proposed finite verifier has not been used here to enlarge the kernel-verified zero-free region;
- the 2026 AI breakthroughs discussed in the public record are motivational context only and are not mathematical inputs.

1.3 Status convention

Tag	Meaning
[proved here]	A complete conventional proof is given in this report.
[imported from Report IV]	Used from the attached unified report and not reproved here.
[conditional on a counterexample]	Deduced under the assumption that a nonintegral solution exists.
[proposed computation]	A concrete exact computation or implementation is proposed, but no output is asserted as a theorem.
[open target]	A precisely stated missing theorem or quantitative target.

2 Scope, nonduplication, and the 2026 trigger

2.1 The inherited frontier

The attached Report IV is approximately twenty-four thousand lines of LaTeX. It already contains the elementary and transcendence reductions, kernel-verified finite exclusions, the canonical primitive generator, exact block counts, radical and Kummer reformulations, auxiliary base spectra, finite differences, generalized Vandermonde determinants, p -adic and tropical ceilings, compact-orbit dynamics, $\mathfrak{o}$ -minimal counting, Loewner and Padé certificates, Mahler systems, cyclotomic and Frobenius constructions, factorial cocycles, moment masks, Bhargava factorials, and a detailed formalization inventory. Repeating any of those developments would make this report longer without moving the frontier.

Accordingly, only one interface theorem is imported: if the conjecture fails, all solutions form the free additive monoid generated by 1 and a unique least noninteger solution β . Everything after Section 3 is organized around a method absent from Report IV: exact algebraic trapping of the power curve by short vectors in a Chebyshev lattice.

2.2 Why the June 2026 paper matters

Brisebarre and Hanrot introduced an algorithmic Coppersmith-style method for locating integer points near a transcendental analytic curve, with Chebyshev interpolation controlling the analytic error and lattice reduction producing auxiliary integer polynomials [3]. Their general algorithm is heuristic at one explicit point: two short auxiliary polynomials can have a common factor, so a resultant computation may fail to isolate the points.

The Alaoglu–Erdős curve has extra structure that removes that difficulty. A polynomial restricted to $(2^t, 3^t)$ becomes an exponential polynomial with pairwise distinct real frequencies. Such functions are strictly totally positive, so a single nonzero auxiliary polynomial has a sharp zero bound, and several independent polynomials give an exact rank–codimension bound. The general Coppersmith machinery is therefore not merely applicable; it simplifies in a way specific to multiplicatively independent bases.

A second 2026 input, due to Wang and Xiao, develops degeneracy theorems for families of unit

equations via integral points and new gcd estimates for exponential Diophantine problems [14]. We use it only to motivate the stable-factor program of Section 12; none of the unconditional results depends on it.

2.3 Competitive context is not evidence

The public 2026 record does substantiate striking AI-assisted mathematics, including an explicit three-dimensional counterexample to the Jacobian conjecture discussed in detail by Tao, and reported constructions covering previously unresolved Hadamard orders below 2000 [11, 12]. Those events justify urgency, not lowered standards. Every claimed advance in this report is proved independently of model attribution, and every remaining gap is stated rather than hidden.

3 The imported conditional interface

Set

$$\mathcal{S} := \{x \in \mathbb{R} : 2^x \in \mathbb{N}, 3^x \in \mathbb{N}\}.$$

Conjecture 3.1 (Alaoglu–Erdős). *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

The report uses the following exact conditional package.

Theorem 3.2 (Primitive solution monoid; imported). *[imported from Report IV] If Conjecture 3.1 is false, there is a unique least noninteger $\beta \in \mathcal{S}$ and integers*

$$M = 2^\beta, \quad A = 3^\beta$$

such that

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\},$$

and every representation is unique. Consequently

$$(2^{n+k\beta}, 3^{n+k\beta}) = (2^n M^k, 3^n A^k).$$

The attached report also gives the kernel-level finite exclusion $\beta \geq 14$ and a software-level exclusion $\beta > 27$.

We shall use the exact unit-block count, also imported:

$$\#(\mathcal{S} \cap [H, H+1)) = 1 + \left\lfloor \frac{H+1}{\beta} \right\rfloor \quad (H \in \mathbb{N}_0). \tag{1}$$

The 1 is the integral solution H .

3.1 Normalized block points

Fix an integer $H \geq 0$ and define

$$K_H := \left\lfloor \frac{H+1}{\beta} \right\rfloor.$$

For $1 \leq k \leq K_H$, the unique solution in the block with second coordinate k is

$$z_{H,k} = \lceil H - k\beta \rceil + k\beta = H + \{k\beta\}.$$

Its integer output pair is

$$X_{H,k} = 2^{z_{H,k}} = 2^{H - \lfloor k\beta \rfloor} M^k, \quad (2)$$

$$Y_{H,k} = 3^{z_{H,k}} = 3^{H - \lfloor k\beta \rfloor} A^k. \quad (3)$$

After dividing by the block scales,

$$\left(\frac{X_{H,k}}{2^H}, \frac{Y_{H,k}}{3^H} \right) = \left(2^{\{k\beta\}}, 3^{\{k\beta\}} \right). \quad (4)$$

Thus the normalized block is the initial orbit segment

$$\mathcal{C}_H = \{(1, 1)\} \cup \left\{ \left(2^{\{k\beta\}}, 3^{\{k\beta\}} \right) : 1 \leq k \leq K_H \right\} \subset \{(2^t, 3^t) : 0 \leq t < 1\}. \quad (5)$$

The sets are nested: $\mathcal{C}_H \subseteq \mathcal{C}_{H+1}$, and at most one new nonintegral point is inserted at each step.

3.2 The exact rational-grid formulation

Put

$$\vartheta := \frac{\log 3}{\log 2}.$$

For $X = 2^{H+t}$ and $Y = 3^{H+t}$ one has

$$\frac{Y}{3^H} = \left(\frac{X}{2^H} \right)^\vartheta.$$

Therefore every exact point in the H th block is an integer pair (X, Y) satisfying

$$2^H \leq X < 2^{H+1}, \quad \frac{Y}{3^H} = f\left(\frac{X}{2^H}\right), \quad f(u) = u^\vartheta, \quad 1 \leq u < 2. \quad (6)$$

This is the Brisebarre–Hanrot rational-grid problem with fixed denominators $u = 2^H$, $v = 3^H$ and zero vertical error. Exactness is crucial: an auxiliary polynomial $P \in \mathbb{Z}[X, Y]$ satisfying

$$\sup_{0 \leq t \leq 1} |P(2^{H+t}, 3^{H+t})| < 1$$

must vanish at every block point because its value there is an integer.

Status: [\[conditional on a counterexample\]](#)

Equations (2)–(5) use the imported primitive-generator theorem. The rational-grid identity (6) is unconditional and applies to every exact integer point of the power curve, whether or not a primitive generator exists.

4 Sparse total positivity on the power curve

Let $E \subset \mathbb{N}_0^2$ be finite. Write

$$q := |E|, \quad \lambda_{i,j} := i \log 2 + j \log 3.$$

The frequencies $\lambda_{i,j}$ are pairwise distinct: equality of two frequencies would give $2^{i-i'}3^{j-j'} = 1$, hence $i = i'$ and $j = j'$.

4.1 The exponential Chebyshev system

Theorem 4.1 (Strict total positivity). *Let*

$$\lambda_1 < \cdots < \lambda_q, \quad t_1 < \cdots < t_m.$$

The matrix

$$V = (e^{\lambda_j t_i})_{1 \leq i \leq m, 1 \leq j \leq q}$$

has rank $\min(m, q)$. More strongly, every minor obtained from increasing row and column indices is strictly positive.

Proof. It is enough to prove that every square generalized Vandermonde determinant

$$\det(e^{\mu_j s_i})_{1 \leq i, j \leq r}$$

is positive for $s_1 < \cdots < s_r$ and $\mu_1 < \cdots < \mu_r$. Divide column j by $e^{\mu_1 s_j}$ after transposing if desired; equivalently, use the classical fact that the Wronskian of $e^{\mu_1 t}, \dots, e^{\mu_r t}$ is

$$e^{(\mu_1 + \cdots + \mu_r)t} \prod_{1 \leq i < j \leq r} (\mu_j - \mu_i) > 0.$$

A family of C^{r-1} functions whose initial Wronskians are everywhere positive is an extended complete Chebyshev system. Its collocation determinants at increasing nodes are positive. For completeness, this last implication can be proved inductively: divide a nontrivial linear combination by the first exponential, differentiate, and apply Rolle's theorem. It follows that no nontrivial combination of r functions can have r real zeros counted with multiplicity; expansion of a collocation determinant along a hypothetical linear dependence would contradict that zero bound. The sign is fixed by coalescing the nodes, where the determinant tends to the positive Wronskian times the positive confluent Vandermonde factor. $\square$

Corollary 4.2 (Sparse zero bound). *Let*

$$P(X, Y) = \sum_{(i,j) \in E} c_{i,j} X^i Y^j \in \mathbb{R}[X, Y]$$

be nonzero. Then

$$F_P(t) := P(2^t, 3^t) = \sum_{(i,j) \in E} c_{i,j} e^{\lambda_{i,j} t}$$

has at most $q - 1$ real zeros, counted with multiplicity.

Proof. Order the distinct frequencies and apply the extended Chebyshev property from Theorem 4.1. $\square$

Remark 4.3 (Why this is stronger than Bézout here). Bézout controls common zeros of two coprime algebraic curves. Corollary 4.2 controls the intersection of a *single* sparse polynomial with the transcendental power curve and remains valid even if several auxiliary polynomials share factors. The relevant complexity is the number of occupied monomials, not the total algebraic degree.

4.2 Multiplicity and evaluation rank

For distinct real nodes $t_1, \dots, t_m$, let

$$\text{ev}_{\mathbf{t}} : \mathbb{R}^E \longrightarrow \mathbb{R}^m, \quad (c_e)_{e \in E} \longmapsto \left(\sum_{e \in E} c_e e^{\lambda_e t_a} \right)_{a=1}^m.$$

Corollary 4.4 (Exact evaluation rank). *For distinct nodes, $\text{rank}(\text{ev}_{\mathbf{t}}) = \min(m, q)$. Hence, when $m < q$,*

$$\dim \ker(\text{ev}_{\mathbf{t}}) = q - m.$$

This elementary dimension identity is the engine behind the deterministic trap theorem.

5 Rank–codimension algebraic traps

5.1 A deterministic replacement for coprimality

Theorem 5.1 (Rank–codimension trap). *Fix a finite support $E \subset \mathbb{N}_0^2$ of size q , a real interval J , and a scale H . Suppose*

$$P_1, \dots, P_r \in \mathbb{Z}[X, Y]$$

have coefficient vectors supported in E , those vectors are linearly independent over $\mathbb{R}$, and

$$\sup_{t \in J} |P_a(2^{H+t}, 3^{H+t})| < 1 \quad (1 \leq a \leq r). \quad (7)$$

Then the curve segment

$$\{(2^{H+t}, 3^{H+t}) : t \in J\}$$

contains at most $q - r$ points of $\mathbb{Z}^2$.

Proof. Let $t_1, \dots, t_m \in J$ be the parameters of distinct integer points. Since each $P_a(2^{H+t_b}, 3^{H+t_b})$ is an integer of absolute value less than one, it is zero. Thus the r coefficient vectors lie in the kernel of the evaluation map at the m nodes. By Corollary 4.4, this kernel has dimension $q - m$ (and is zero if $m \geq q$). Therefore $r \leq q - m$, or $m \leq q - r$. $\square$

Status: [\[proved here\]](#)

Theorem 5.1 is exact and deterministic. It needs neither a resultant nor a coprimality test. The only numerical statement to certify is the strict bound (7); the rank of the integer coefficient vectors is checked exactly.

Corollary 5.2 (One polynomial). *A single nonzero q -monomial polynomial satisfying (7) bounds the number of exact integer points by $q - 1$.*

Corollary 5.3 (Several supports). *Suppose $P_1, \dots, P_r$ have possibly different supports. Let E be their union and let r' be the rank of their coefficient vectors after padding by zero in $\mathbb{R}^E$. Then the same segment contains at most $|E| - r'$ integer points.*

5.2 Piecewise certificates

Global uniform approximation is not mandatory. Let $J_1, \dots, J_s$ be intervals with disjoint interiors covering J . On J_ν , let E_ν be a support of size q_ν and let r_ν be the rank of certified short polynomials.

Criterion 5.4 (Cover certificate). *The number of exact integer points with parameter in J is at most*

$$\sum_{\nu=1}^s (q_\nu - r_\nu), \quad (8)$$

with endpoints assigned to one neighboring interval to avoid double counting.

The quantity $q_\nu - r_\nu$ is the *local coefficient defect*. Formula (8) turns the search for points into a search for low total defect.

Corollary 5.5 (Conditional closing criterion). *Assume a counterexample exists with primitive generator β . If, for infinitely many integers H , one can cover $[0, 1]$ by certified traps whose total defect is $o(H)$, then Conjecture 3.1 follows.*

Proof. The exact block count (1) is $H/\beta + O(1)$, while the cover certificate gives $o(H)$. □

5.3 Successive minima are the correct invariant

Every analytic construction below associates to (H, J, E) a lattice $\mathcal{L}_{H,J,E}$ whose short vectors are coefficient vectors of uniformly small integer polynomials. Let

$$0 < \mu_1 \leq \dots \leq \mu_q$$

be its successive minima relative to the norm that dominates the curve sup norm, and let τ be the strict threshold corresponding to “supremum < 1 .”

Criterion 5.6 (Successive-minimum point bound). *If $\mu_r < \tau$, then the interval contains at most $q - r$ exact integer points. Conversely, if the interval contains m exact integer points, then*

$$\mu_{q-m+1} \geq \tau.$$

Thus the problem is spectral. Finding the first short vector proves a $q - 1$ bound; closing the conjecture requires control near the *top* of the successive-minimum spectrum.

6 A one-dimensional Chebyshev lattice on the exact curve

The general Brisebarre–Hanrot construction approximates a two-dimensional strip around a curve. Exact equality permits a leaner object: expand each monomial directly along

$$\Phi_H(t) = (2^{H+t}, 3^{H+t}), \quad 0 \leq t \leq 1.$$

This reduces the analytic part to one variable and exposes the frequency geometry.

6.1 Bernstein ellipses and coefficient bounds

Map $t \in [0, 1]$ to $u = 2t - 1 \in [-1, 1]$. For $\rho > 1$, let $\mathcal{E}_\rho$ be the Bernstein ellipse in the u -plane and put

$$R(\rho) := \frac{1}{2} + \frac{\rho + \rho^{-1}}{4}. \quad (9)$$

The maximum real part of $t = (1 + u)/2$ on $\mathcal{E}_\rho$ is $R(\rho)$.

For $e = (i, j) \in E$ define

$$\lambda_e = i \log 2 + j \log 3, \quad f_e(t) = e^{\lambda_e(H+t)}.$$

Write its Chebyshev expansion

$$f_e(t) = \sum_{n \geq 0} a_{n,e} T_n(2t - 1),$$

where the prime only changes the conventional normalization of the constant coefficient. Standard contour estimates give, after harmless adjustment of the $n = 0$ factor,

$$|a_{n,e}| \leq 2M_e \rho^{-n}, \quad M_e := e^{\lambda_e(H+R(\rho))}. \quad (10)$$

Consequently the tail after q terms is bounded by

$$\sum_{n \geq q} |a_{n,e}| \leq \tau_e, \quad \tau_e := \frac{2M_e \rho^{-q}}{1 - \rho^{-1}}. \quad (11)$$

6.2 The coefficient-and-tail matrix

Let A be the $q \times q$ real matrix with entries $A_{n,e} = a_{n,e}$, $0 \leq n < q$, and let $D = \text{diag}(\tau_e)$. Define the $2q \times q$ matrix

$$B = B(H, E, \rho) := \begin{pmatrix} A \\ D \end{pmatrix} \quad (12)$$

and the rank- q lattice

$$\mathcal{L}(H, E, \rho) := B\mathbb{Z}^E \subset \mathbb{R}^{2q}.$$

For $c = (c_e) \in \mathbb{Z}^E$, let

$$P_c(X, Y) = \sum_{e=(i,j) \in E} c_e X^i Y^j.$$

Lemma 6.1 (Norm-to-supremum transfer). *For every $c \in \mathbb{Z}^E$,*

$$\sup_{0 \leq t \leq 1} |P_c(2^{H+t}, 3^{H+t})| \leq \sqrt{2q} \|Bc\|_2. \quad (13)$$

Proof. The first q Chebyshev coefficients of the linear combination are Ac . Their absolute sum is at most $\sqrt{q} \|Ac\|_2$. By (11), the tail is at most $\sum_e |c_e| \tau_e \leq \sqrt{q} \|Dc\|_2$. Since $|T_n(u)| \leq 1$ on $[-1, 1]$, the sum of these bounds is at most $\sqrt{2q} \|Bc\|_2$. $\square$

Thus a lattice vector shorter than $(2q)^{-1/2}$ is a valid algebraic trap.

6.3 Covolume by Cauchy–Binet

Let

$$\Delta(H, E, \rho) := \sqrt{\det(B^T B)}$$

be the covolume of $\mathcal{L}(H, E, \rho)$ in its span.

Lemma 6.2 (Universal covolume bound). *Assume $\rho \geq 2$. Uniformly over all finite supports E of size q ,*

$$\log \Delta(H, E, \rho) \leq (H + R(\rho)) \sum_{e \in E} \lambda_e - \frac{q(q-1)}{2} \log \rho + O(q \log q), \quad (14)$$

where the implied constant is absolute.

Proof. Cauchy–Binet writes $\det(B^T B)$ as the sum of squares of all $q \times q$ minors of B . Choose k rows from A and $q - k$ diagonal rows from D . A nonzero minor factors into a $k \times k$ coefficient minor and $q - k$ diagonal weights. By (10), Hadamard’s inequality bounds its absolute value by

$$e^{(H+R) \sum_{e \in E} \lambda_e} \rho^{-\sum_{n \in S} n - q(q-k)} e^{O(q \log q)},$$

where S is the chosen k -set of Chebyshev indices. The smallest possible exponent of ρ^{-1} is obtained from $S = \{0, \dots, k-1\}$, and

$$\begin{aligned} q(q-k) + \frac{k(k-1)}{2} &= \frac{q(q-1)}{2} + \frac{(q-k)(q-k+1)}{2} \\ &\geq \frac{q(q-1)}{2}. \end{aligned}$$

The number of minors and all factors 2, $(1 - \rho^{-1})^{-1}$, and Hadamard constants contribute $e^{O(q \log q)}$. Taking the square root of the Cauchy–Binet sum gives (14). $\square$

The identity in the proof is important: diagonal tail rows never reduce the full $q(q-1)/2$ Chebyshev decay. They only add further decay.

6.4 Minkowski and fixed-rank short vectors

Let $\mu_1 \leq \dots \leq \mu_q$ be the Euclidean successive minima of $\mathcal{L}(H, E, \rho)$. Minkowski's second theorem gives

$$\mu_1 \cdots \mu_q \operatorname{vol}(B_2^q) \leq 2^q \Delta(H, E, \rho).$$

In particular,

$$\log \mu_1 \leq \frac{H + R(\rho)}{q} \sum_{e \in E} \lambda_e - \frac{q-1}{2} \log \rho + O(\log q). \quad (15)$$

To obtain several vectors, note that every nonzero $c \in \mathbb{Z}^E$ obeys

$$\|Bc\|_2 \geq \min_e \tau_e.$$

When $(0, 0) \in E$, the minimum is at least $2\rho^{-q}$ for $\rho \geq 2$. Therefore, for each fixed r ,

$$\log \mu_r \leq \frac{H + R(\rho)}{q} \sum_{e \in E} \lambda_e - \frac{q}{2} \log \rho + O_r(\log q + \log \rho). \quad (16)$$

The difference between $q/2$ and $(q-1)/2$ is absorbed in the error.

Theorem 6.3 (Fixed-rank Chebyshev trap). *Let r be fixed. Suppose $(0, 0) \in E_H$, $q_H = |E_H| \rightarrow \infty$, and there are parameters $\rho_H \geq 2$ such that*

$$\frac{H + R(\rho_H)}{q_H} \sum_{e \in E_H} \lambda_e - \frac{q_H}{2} \log \rho_H \rightarrow -\infty \quad (17)$$

faster than $\log q_H + \log \rho_H$. Then, for all sufficiently large H , there are r linearly independent polynomials in $\mathbb{Z}[X, Y]$, supported in E_H , whose absolute values on $\Phi_H([0, 1])$ are strictly less than one.

Proof. Equation (16) makes $\mu_r < (2q_H)^{-1/2}$ for large H . Choose r independent lattice vectors below that threshold and apply Lemma 6.1. $\square$

Status: [\[proved here\]](#)

Theorem 6.3 is an existence theorem in the exact real Chebyshev lattice. A proof-producing implementation replaces the real coefficients by outward-rounded rational intervals and verifies the final sup-norm inequality independently; see Section 16.

7 Frequency-simplex optimization and a new block bound

The choice of monomial support determines both sides of (17). For fixed cardinality q , the negative Chebyshev term depends only on q , while the positive growth term is $\sum_{e \in E} \lambda_e$. The optimal support is therefore obtained by taking the q smallest frequencies.

Put

$$a := \log 2, \quad b := \log 3.$$

For $L > 0$, define the weighted frequency simplex

$$E_L := \{(i, j) \in \mathbb{N}_0^2 : ai + bj \leq L\}. \quad (18)$$

Let

$$Q(L) := |E_L|, \quad S(L) := \sum_{(i,j) \in E_L} (ai + bj).$$

7.1 Lattice-point asymptotics

Lemma 7.1 (Weighted simplex count and first moment). *As $L \rightarrow \infty$,*

$$Q(L) = \frac{L^2}{2ab} + O(L), \quad (19)$$

$$S(L) = \frac{L^3}{3ab} + O(L^2), \quad (20)$$

$$\frac{S(L)}{Q(L)} = \frac{2L}{3} + O(1). \quad (21)$$

Proof. Summing vertical fibers gives

$$Q(L) = \sum_{0 \leq i \leq L/a} \left(1 + \left\lfloor \frac{L - ai}{b} \right\rfloor\right).$$

Replacing each floor by its argument incurs $O(L)$, and the remaining arithmetic progression is the area $L^2/(2ab)$ plus $O(L)$. For the first moment, sum

$$\sum_i \sum_{0 \leq j \leq (L - ai)/b} (ai + bj).$$

Replacing the inner sum by its integral incurs $O(L^2)$. The integral over the triangle $ai + bj \leq L$ is

$$\int_0^{L/a} \int_0^{(L - ai)/b} (ai + bj) \, dj \, di = \frac{L^3}{3ab}.$$

Division yields (21). □

Proposition 7.2 (Asymptotic support optimality). *Among all q -element subsets of $\mathbb{N}_0^2$, the sum of frequencies is minimized by the q lattice points with smallest values of $ai + bj$. Hence weighted simplices are asymptotically optimal for the Chebyshev covolume problem.*

Proof. This is the rearrangement principle: if a support contains a frequency larger than an omitted one, swapping the two decreases the sum without changing q . □

The usual total-degree triangle $i + j \leq d$ has average frequency $d(a + b)/3$. At the same cardinality, the weighted simplex has asymptotic average $2\sqrt{2abq}/3$, improving the leading constant by the factor

$$\frac{2\sqrt{ab}}{a + b} = 0.973 \dots$$

The gain is modest but exact and costs nothing.

7.2 Choice of the Bernstein ellipse

Insert E_L into (15). By Lemma 7.1,

$$\log \mu_1 \leq \frac{2L}{3}(H + R(\rho)) - \frac{L^2}{4ab} \log \rho + O(L^2 + \log L). \quad (22)$$

The $O(L^2)$ term absorbs the boundary errors from $Q(L)$ and $S(L)$.

For the unit block, the optimal ellipse grows with L . Since $R(\rho) = \rho/4 + 1/2 + O(\rho^{-1})$, differentiating the two ρ -dependent leading terms suggests

$$\rho = \frac{3L}{2ab}. \quad (23)$$

With this choice,

$$\log \mu_1 \leq \frac{2LH}{3} - \frac{L^2}{4ab} \log L + O(L^2). \quad (24)$$

The same leading estimate holds for every fixed successive minimum μ_r by (16).

7.3 The logarithmic-square saving

Theorem 7.3 (Unconditional integer-point bound in a logarithmic unit block). *Let*

$$I(H) := \#\{x \in [H, H+1) : 2^x \in \mathbb{Z}, 3^x \in \mathbb{Z}\}.$$

Then

$$\boxed{\limsup_{H \rightarrow \infty} I(H) \frac{(\log H)^2}{H^2} \leq \frac{32 \log 2 \log 3}{9}} \quad (25)$$

and

$$\frac{32 \log 2 \log 3}{9} = 2.70755559260021 \dots$$

More precisely, for every $\varepsilon > 0$ and every fixed $r \geq 1$, all sufficiently large H admit r independent uniformly short integer polynomials supported on a set of size

$$\left(\frac{32 \log 2 \log 3}{9} + \varepsilon + o(1) \right) \frac{H^2}{(\log H)^2}, \quad (26)$$

so Theorem 5.1 subtracts r from that bound.

Proof. Fix $\delta > 0$ and choose

$$L = \left(\frac{8ab}{3} + \delta \right) \frac{H}{\log H}, \quad \rho = \frac{3L}{2ab}. \quad (27)$$

Since $\log L = \log H + O(\log \log H)$, the two leading terms in (24) equal

$$\frac{2LH}{3} - \frac{L^2}{4ab} \log L = -c_\delta \frac{H^2}{\log H} + O\left(\frac{H^2 \log \log H}{(\log H)^2} \right)$$

for some $c_\delta > 0$. Thus the fixed-rank shortness criterion (17) holds with a margin tending to $-\infty$. Theorem 6.3 and then Theorem 5.1 give $I(H) \leq Q(L) - r$.

Finally,

$$Q(L) = \frac{L^2}{2ab} + O(L) = \left(\frac{32ab}{9} + O(\delta) \right) \frac{H^2}{(\log H)^2}.$$

Let $\delta \downarrow 0$. □

Status: [\[proved here\]](#)

Theorem 7.3 is the principal unconditional advance of this report. It is not a reformulation of the conjecture and does not assume the primitive solution monoid. Its input is only multiplicative independence of 2 and 3, Chebyshev coefficient decay, Cauchy–Binet, Minkowski’s theorem, and strict total positivity.

7.4 Why the theorem still falls short

Under a counterexample, (1) gives

$$I(H) = \frac{H}{\beta} + O(1).$$

The new upper bound is

$$I(H) \ll \frac{H^2}{(\log H)^2},$$

which is smaller than a quadratic bound by a logarithmic square but still larger than the conditional linear count by the factor $H/(\log H)^2$. This is not a matter of constants. A proof needs either:

- a much smaller coefficient support;
- a rank r so close to q that the defect $q - r$ becomes $o(H)$; or
- an additional arithmetic mechanism that forces multiplicity or shared factors without paying the full coefficient-space dimension.

The next sections quantify the second possibility and audit the third.

8 Macroscopic windows and the exact high-corank threshold

The unit-block theorem uses an ellipse whose radius grows like $H/\log H$. To compare the trap dimension directly with the quadratic cumulative population forced by a counterexample, consider a macroscopic exponent window

$$\mathcal{I}_{H,r} := [H, (1+r)H], \quad r > 0. \tag{28}$$

Parameterize it by $z = H(1 + rs)$, $0 \leq s \leq 1$.

8.1 Trap dimension in a fixed-ratio window

For a weighted simplex with q monomials, its average frequency is

$$\frac{1}{q} \sum_{e \in E} \lambda_e = \frac{2}{3} \sqrt{2abq} + o(\sqrt{q}). \tag{29}$$

On a Bernstein ellipse of fixed parameter $\rho > 1$, the largest real exponent is $H(1 + rR(\rho))$. The first-minimum inequality becomes negative when

$$\sqrt{q} > \frac{4\sqrt{2ab}}{3} \frac{1 + rR(\rho)}{\log \rho} H + o(H). \quad (30)$$

Thus a fixed-rank trap can be built with

$$q = \left(\frac{32ab}{9} \frac{(1 + rR(\rho))^2}{(\log \rho)^2} + o(1) \right) H^2. \quad (31)$$

8.2 Population supplied by a counterexample

The cumulative conditional count below exponent T is $T^2/(2\beta) + O(T)$. Therefore the window (28) contains

$$\left(\frac{r + r^2/2}{\beta} + o(1) \right) H^2 \quad (32)$$

solutions. The ratio between the fixed-rank trap dimension and this population is asymptotically

$$\beta \kappa(r, \rho), \quad \kappa(r, \rho) := \frac{32ab}{9} \frac{(1 + rR(\rho))^2}{(r + r^2/2)(\log \rho)^2}. \quad (33)$$

8.3 Exact optimization

Theorem 8.1 (Optimal macroscopic-window constant). *Let $t_0 > 0$ be the unique solution of*

$$t \tanh t = 2. \quad (34)$$

Then

$$\kappa_* := \inf_{r>0, \rho>1} \kappa(r, \rho) = \frac{64 \log 2 \log 3 \cosh t_0}{9 t_0^2} = 5.08726658072721 \dots \quad (35)$$

The minimizers are

$$t_0 = 2.06533813897470 \dots, \quad (36)$$

$$\rho_0 = e^{t_0} = 7.88796467504653 \dots, \quad (37)$$

$$r_0 = 0.665032892479158 \dots \quad (38)$$

Proof. For fixed ρ , write $R = R(\rho)$. Differentiating $(1 + rR)^2/(r + r^2/2)$ gives the unique minimum

$$r = \frac{1}{R - 1}.$$

At this value,

$$\frac{(1 + rR)^2}{r + r^2/2} = 2(2R - 1).$$

Put $t = \log \rho$. Since

$$2R(\rho) - 1 = \frac{\rho + \rho^{-1}}{2} = \cosh t,$$

we obtain

$$\inf_r \kappa(r, e^t) = \frac{64ab \cosh t}{9 t^2}.$$

The derivative vanishes exactly when $t \sinh t = 2 \cosh t$, or $t \tanh t = 2$. The left side is strictly increasing from 0 to infinity, so the root is unique. Substitution gives the numerical values. $\square$

8.4 The required rank fraction

If a trap space has dimension q and certified short rank R_H , its point bound is $q - R_H$. In the optimized macroscopic window, contradiction with (32) requires

$$\frac{R_H}{q} > 1 - \frac{1}{\kappa_* \beta} + o(1). \quad (39)$$

Corollary 8.2 (Numerical high-corank frontier). *Using the imported kernel-level bound $\beta \geq 14$, a closing trap must certify*

$$\frac{R_H}{q} > 0.985959341761414\dots, \quad (40)$$

that is, more than 98.59% of all coefficient directions. Using the existing software-level exclusion $\beta > 27$, the threshold becomes

$$\frac{R_H}{q} > 0.992719658691104\dots \quad (41)$$

Status: [\[proved here\]](#)

The constants in Theorem 8.1 and Corollary 8.2 are analytic, not fitted numerically. The only numerical operation is evaluation of the unique root of $t \tanh t = 2$.

8.5 Interpretation

A two-polynomial Coppersmith calculation is nowhere near the required regime. Even a theorem producing ten, one hundred, or a fixed positive number of independent short vectors would not change the asymptotic comparison. The needed statement is qualitatively different: the lattice must have only about 1.4% of its directions above threshold at the kernel frontier, and less than 0.73% at the software frontier. This is why the next natural language is parametric geometry of numbers rather than ordinary LLL output.

9 What determinant volume can and cannot say about rank

Minkowski’s second theorem controls the product of all successive minima. Turning that product into a bound for a high individual minimum requires lower bounds for the earlier minima. The diagonal tail rows provide such a lower bound, but it is exponentially small and therefore expensive when used repeatedly.

9.1 A general rank-tax inequality

Assume $(0, 0) \in E$ and $\rho \geq 2$. From (11), every nonzero lattice vector has norm at least $2\rho^{-q}$. If $1 \leq r \leq q$, then

$$\mu_1 \cdots \mu_q \geq (2\rho^{-q})^{r-1} \mu_r^{q-r+1}.$$

Combining this with Minkowski’s second theorem and Lemma 6.2 yields

$$(q - r + 1) \log \mu_r \leq (H + R(\rho)) \sum_{e \in E} \lambda_e - \frac{q(q-1)}{2} \log \rho$$

$$+ (r - 1)q \log \rho + O(q \log q). \quad (42)$$

The term $(r - 1)q \log \rho$ is the *rank tax*. It is absent for the first minimum and lower order for fixed r , but it is quadratic in q when r is a positive fraction of the dimension.

9.2 The one-half ceiling of this proof mechanism

Ignore the positive archimedean term in (42); this only makes short vectors easier. If $r = \alpha q + o(q)$, the leading $\log \rho$ coefficient in the numerator is

$$\left(-\frac{1}{2} + \alpha\right) q^2.$$

Therefore this determinant-plus-universal-lower-bound argument cannot prove $\mu_r < 1$ for any fixed $\alpha > 1/2$, even in the fictitious case where all monomial growth vanishes. This is a ceiling of the *argument*, not a lower bound on the actual minima. It shows that the 98.59% target in (40) cannot be reached by feeding the same covolume estimate into Minkowski more aggressively.

Caution 9.1 (Volume is not spectrum). A small determinant can coexist with a few extremely short directions and many long ones. Conversely, near-full short rank is a statement about the shape of the lattice, not merely its covolume. Any future claim that “the determinant is tiny, hence almost every basis vector is short” must supply a genuine condition-number, transference, or successive-minimum theorem.

9.3 A conservation law in the unit-block asymptotic

For the unit-block weighted simplex, let the outer frequency scale be $L = cH/\log H$. The determinant margin permits, through (42), a rank fraction no larger than approximately

$$\alpha(c) < \frac{1}{2} - \frac{4ab}{3c}, \quad c > \frac{8ab}{3}.$$

The residual coefficient defect has leading size

$$(1 - \alpha(c))Q(L) \gtrsim \left(\frac{1}{2} + \frac{4ab}{3c}\right) \frac{c^2}{2ab} \frac{H^2}{(\log H)^2}.$$

The right side is minimized at the threshold $c = 8ab/3$, where $\alpha = 0$, and reproduces exactly $32ab/9$. Increasing the support makes more rank available, but the larger ambient dimension cancels the gain. Within this coarse volume method, fixed-rank and fractional-rank optimization have the same leading point-count constant.

10 Parametric geometry of the trap lattices

The scale H enters each monomial column through

$$e^{\lambda_e H}.$$

After separating the H -independent Chebyshev data, the family is a diagonal flow with weight vector $(\lambda_e)_{e \in E}$. This is the natural setting of Schmidt–Summerer parametric geometry of numbers and Roy’s rigid-system realization theory [9, 10, 8].

10.1 Successive-minimum functions

For a fixed support and a fixed normalized interval, write

$$L_i(H) := \log \mu_i(H), \quad 1 \leq i \leq q.$$

The columns of the defining lattice are multiplied by $e^{\lambda_e H}$ as H varies. Standard parametric-geometry heuristics predict piecewise-linear behavior for the ordered functions $L_i(H)$, with slopes drawn from finite sums of the weights λ_e and switch points at which distinct short-vector supports exchange order.

Two features make this family more structured than a generic diagonal flow:

1. the weights are the two-dimensional semigroup $i \log 2 + j \log 3$;
2. the analytic rows are Chebyshev coefficients of the same exponentials, so their H -independent part is strictly totally positive.

A high-corank theorem would follow from a strong upper bound on the upper tail of $L_i(H)$, not from control of their sum alone.

Research target 10.1 (Trap-spectrum theorem). *For weighted-simplex supports in the optimized macroscopic window, prove one of the following:*

1. *at least $(1 - 1/(\kappa_*\beta) + \eta)q$ successive minima lie below the exact trap threshold for infinitely many H ;*
2. *the set of indices above threshold has cardinality $o(q)$;*
3. *a transference inequality bounds the long tail by arithmetic data at primes dividing $6MA$ and forces the same conclusion under a hypothetical primitive pair (M, A) .*

Any one of these statements, with the first using $\beta \geq 14$, proves the conjecture.

10.2 Switch density as a finite combinatorial target

The coefficient vector of a short polynomial has a support pattern, sign pattern, and a leading Chebyshev cancellation pattern. If only $o(q)$ directions are long, then almost all patterns must persist over substantial H -intervals. Conversely, if patterns switch too often, pairs of nearly equal exponential weights arise. Their gaps are integer linear forms in $\log 2$ and $\log 3$:

$$|(i - i') \log 2 + (j - j') \log 3|.$$

Baker's theorem gives explicit lower bounds for nonzero gaps. This suggests a concrete program:

Research target 10.2 (Switch-count bound). *Show that the number of changes in the ordered short-vector support data over an H -interval of length T is $o(qT)$, with an explicit defect small enough to force either a stable subspace or a contradiction with the rank threshold.*

This target uses Baker theory in a role different from direct approximation: not to exclude a single counterexample, but to limit how rapidly the lattice spectrum can reorganize.

11 Dilation rigidity and multiscale persistence

The normalized orbit sets are nested, while unnormalized points evolve by

$$(X, Y) \mapsto (2X, 3Y).$$

Exact reuse of an auxiliary polynomial across scales is severely rigid.

11.1 A sparse dilation theorem

Theorem 11.1 (Consecutive-dilation rigidity). *Let*

$$P(X, Y) = \sum_{e=(i,j) \in E} c_e X^i Y^j$$

have q nonzero monomials, and let $X, Y > 0$. If

$$P(2^n X, 3^n Y) = 0 \quad (n = n_0, n_0 + 1, \dots, n_0 + q - 1), \quad (43)$$

then $P = 0$.

Proof. Equation (43) is the square linear system

$$\sum_{e=(i,j) \in E} (c_e X^i Y^j) (2^i 3^j)^n = 0.$$

The bases $2^i 3^j$ are distinct. After factoring their n_0 th powers, the coefficient matrix is an ordinary Vandermonde matrix and is invertible. Hence every $c_e X^i Y^j$ is zero, so every c_e is zero. $\square$

Corollary 11.2 (No fixed trap for a persistent orbit point). *A nonzero q -monomial polynomial cannot vanish at q consecutive unnormalized incarnations of the same normalized orbit point.*

This matters because a fixed orbit index k persists: once $k \leq K_H$,

$$(X_{H+1,k}, Y_{H+1,k}) = (2X_{H,k}, 3Y_{H,k}).$$

A multiscale proof cannot simply discover one polynomial and replay it indefinitely.

11.2 Invariant subspaces of the dilation operator

Define

$$\mathcal{D}(P)(X, Y) := P(2X, 3Y).$$

On monomials,

$$\mathcal{D}(X^i Y^j) = 2^i 3^j X^i Y^j.$$

The eigenvalues are simple.

Proposition 11.3 (Invariant-space classification). *Every finite-dimensional $\mathcal{D}$ -invariant subspace of $\mathbb{R}[X, Y]$ is spanned by monomials.*

Proof. The operator is diagonal in the monomial basis with pairwise distinct eigenvalues. Every invariant subspace of a diagonal operator with simple spectrum is the span of a subset of its eigenvectors. $\square$

A monomial never vanishes on the positive curve. Therefore an exactly invariant trap space cannot solve the problem.

11.3 The viable replacement: controlled covariance

Exact invariance is too rigid, but adjacent trap lattices are not unrelated. The coefficient weights are multiplied by $2^i 3^j$, while the normalized point set changes by at most one orbit insertion. A useful theorem would permit a low-rank error:

Research target 11.4 (Approximate dilation covariance). *Construct trap subspaces $W_H \subset \mathbb{R}^{E_H}$ and transport maps $T_H : W_H \rightarrow W_{H+1}$ such that*

1. $\dim W_H = (1 - o(1))|E_H|$;
2. *the uniformly short property is preserved by T_H ;*
3. $\dim(W_{H+1}/T_H W_H) = o(H)$;
4. *the accumulated transport defect is incompatible with Theorem 11.1 unless no nonintegral orbit is present.*

The fourth clause is essential: without it, a new basis can be chosen independently at every scale and nesting supplies no leverage.

12 Stable factors and unit-equation degeneration

If two short polynomials share a factor, the general Brisebarre–Hanrot algorithm may return “fail.” On the power curve, common factors are not an obstacle to point counting, but they may contain useful multiscale information.

12.1 Finite recurrence is impossible

Proposition 12.1 (Finite projective recurrence). *Let $V \subset \mathbb{R}[X, Y]$ be finite dimensional. Suppose that for infinitely many H there is a nonzero $P_H \in V$ vanishing on the entire normalized set $\mathcal{C}_H$, and suppose the projective classes $[P_H] \in \mathbb{P}(V)$ take only finitely many values. Then no counterexample exists.*

Proof. One projective class occurs for infinitely many H . Since the sets $\mathcal{C}_H$ are nested and their cardinalities tend to infinity under a counterexample, the corresponding fixed polynomial vanishes at infinitely many points of the power arc. Corollary 4.2 forces it to be zero, a contradiction. $\square$

The proposition remains true when “finitely many values” is replaced by any condition that forces a fixed nonzero limit polynomial and controls convergence strongly enough to pass zeros to the limit.

12.2 Where unit-equation theory may enter

Factor coefficients produced by trap lattices are large integers with structured prime content. If a bounded-degree factor recurs after the dilation rescaling $(X, Y) \mapsto (2X, 3Y)$, its coefficients satisfy exponential recurrences. Relations among several such factors become one-parameter families of unit equations. The 2026 results of Wang–Xiao show that, under transverse boundary-divisor hypotheses, such families are forced into proper algebraic subsets and yield new gcd estimates for exponential Diophantine problems [14].

The missing bridge is explicit:

Research target 12.2 (Stable-factor degeneration). *Show that a common factor occurring in a positive-density set of optimized trap spaces must, after monomial normalization, belong to a fixed algebraic family whose coefficient evolution is governed by a unit equation. Prove that every degenerate family is monomial or binomial. The monomial case has no positive zeros; the binomial case would force a rational relation between $\log 2$ and $\log 3$.*

This route is secondary to the successive-minimum target, but it is the correct response if large common factors repeatedly appear in exact computations.

13 Hermite multiplicity: a tempting route and its row-reset obstruction

Multiplicity appears to offer an immediate improvement. If an integer polynomial and all its partial derivatives of total order below s are uniformly smaller than one, then all those integer values vanish at every exact point. Each point becomes a zero of multiplicity at least s on the restricted exponential polynomial.

Proposition 13.1 (Hermite trap bound). *Let $P \in \mathbb{Z}[X, Y]$ have q nonzero monomials. Suppose that for every $\alpha = (\alpha_1, \alpha_2) \in \mathbb{N}_0^2$ with $|\alpha| < s$,*

$$\sup_{t \in J} \left| \partial_X^{\alpha_1} \partial_Y^{\alpha_2} P(2^{H+t}, 3^{H+t}) \right| < 1. \quad (44)$$

Then the segment contains at most $(q - 1)/s$ exact integer points.

Proof. Every derivative value in (44) is an integer at an integer point, hence zero. Thus P has algebraic multiplicity at least s there. Composing with the analytic curve preserves at least that order of vanishing. Corollary 4.2 bounds the total zero multiplicity by $q - 1$. $\square$

If s could grow like the square root of q at negligible cost, this would reduce a quadratic coefficient budget to a linear point bound and might prove the conjecture. The cost is not negligible.

13.1 Why derivatives are not free

For one derivative family, high-order Chebyshev modes carry the decay ρ^{-n} . Adding all jets of order below s creates

$$J_s = \binom{s+1}{2}$$

new families, each with its own low Chebyshev modes. A Cauchy–Binet minor can replace a very expensive mode $n \asymp q$ by mode 0 from a derivative family. The derivative contributes a scale penalty roughly

$$2^{-H\alpha_1} 3^{-H\alpha_2}$$

but saves a Chebyshev penalty ρ^{-n} . In the relevant regime, $q \log \rho \asymp H^2 / \log H$ is much larger than H , so these row resets dominate the naive entrywise smallness of derivatives.

There is also an exact algebraic calibration.

Lemma 13.2 (Jet matrix nondegeneracy). *Let*

$$\mathcal{T}_s := \{(i, j) \in \mathbb{N}_0^2 : i + j < s\}.$$

At any point (X_0, Y_0) with $X_0 Y_0 \neq 0$, the square matrix

$$\left(\partial_X^{\alpha_1} \partial_Y^{\alpha_2} X^i Y^j \Big|_{(X_0, Y_0)} \right)_{\alpha \in \mathcal{T}_s, (i, j) \in \mathcal{T}_s}$$

is invertible.

Proof. If a polynomial of total degree below s has all jets of total order below s equal to zero at (X_0, Y_0) , its Taylor polynomial there is identically zero; hence the polynomial is zero. Thus the jet evaluation map is injective between vector spaces of the same dimension. $\square$

Lemma 13.2 says that J_s zero-mode jet rows can consume J_s coefficient directions before any Chebyshev decay is used. Multiplicity has a quadratic interpolation tax.

Research target 13.3 (Subcritical Hermite regime). *Determine the largest $s = s(q, H)$ for which the combined jet–Chebyshev lattice still has a short nonzero vector. The interesting range is*

$$\frac{H}{(\log H)^2} \ll s \ll \frac{H}{\log H},$$

where multiplicity would reduce q/s to $o(H)$ but the jet count J_s is still $o(q)$. A positive result in this range would prove the conjecture; a matching negative result would close the most direct multiplicity escape.

The target is concrete: it is a min-plus assignment problem between derivative order and Chebyshev mode, followed by one covolume estimate. No transcendence theorem is hidden in its statement.

14 Exact Bessel factorization of the analytic lattice

The generic Bernstein-ellipse estimate is sufficient for asymptotics, but the present curve has an exact Chebyshev expansion. With $u = 2t - 1$,

$$e^{\lambda(H+t)} = e^{\lambda(H+1/2)} e^{(\lambda/2)u}.$$

The classical generating formula for modified Bessel functions is

$$e^{zu} = I_0(z) + 2 \sum_{n \geq 1} I_n(z) T_n(u). \quad (45)$$

Therefore

$$a_{n,e} = \epsilon_n e^{\lambda_e(H+1/2)} I_n(\lambda_e/2), \quad \epsilon_0 = 1, \quad \epsilon_n = 2 \ (n \geq 1). \quad (46)$$

Proposition 14.1 (Diagonal-flow factorization). *For the first N Chebyshev rows and a support E , the coefficient matrix factors as*

$$A_N(H, E) = D_{\text{row}} K_N(E) D_H, \quad (47)$$

where

$$(D_{\text{row}})_{n,n} = \epsilon_n, \quad K_N(E)_{n,e} = I_n(\lambda_e/2), \quad (D_H)_{e,e} = e^{\lambda_e(H+1/2)}.$$

All dependence on H is in the final diagonal matrix.

This makes the parametric-geometry structure exact rather than heuristic. It also gives a clean finite-verification route: interval enclosures for $\log 2$, $\log 3$, e^x , and $I_n(x)$ produce certified enclosures for every matrix entry, while the Bessel series gives a positive tail bound.

14.1 Compound matrices as the spectral bottleneck

The k th exterior power satisfies

$$\bigwedge^k A_N(H, E) = \left(\bigwedge^k D_{\text{row}}\right) \left(\bigwedge^k K_N(E)\right) \left(\bigwedge^k D_H\right).$$

The last factor is explicit: its diagonal entries are $e^{H \sum_{e \in F} \lambda_e}$ over k -subsets $F \subset E$. Thus every nontrivial condition-number question is concentrated in the fixed Bessel matrix $K_N(E)$.

Research target 14.2 (Bessel compound bounds). *For weighted-simplex frequency sets, obtain uniform lower and upper bounds for a $1 - o(1)$ fraction of the singular values of $K_q(E)$, or equivalently for the bulk of its compound minors. Bounds strong enough to prevent concentration of covolume in $o(q)$ singular directions would convert the determinant estimate into the high-corank conclusion required by Corollary 8.2.*

The entries of K are positive and the underlying exponential kernel is strictly totally positive. This makes total-positivity techniques, oscillatory matrices, and variation- diminishing factorizations natural tools. The report does not assume that the finite Bessel matrix is uniformly well conditioned; establishing or refuting precisely that assertion is the point of Target 14.2.

14.2 An exact tail formula

For $z \geq 0$, every $I_n(z)$ is positive and

$$I_n(z) = \sum_{k \geq 0} \frac{(z/2)^{n+2k}}{k!(n+k)!}.$$

Hence a rational interval upper bound for z gives a monotone, proof-producing upper bound on

$$2e^{\lambda(H+1/2)} \sum_{n \geq N} I_n(\lambda/2).$$

One may bound this series either termwise or by the Bernstein estimate (11). In a finite certificate, the smaller of the two bounds can be stored. No numerical quadrature is required.

15 General multiplicatively independent bases

The method is not special to 2 and 3. Let $u, v > 1$ be multiplicatively independent integers and put

$$a_u = \log u, \quad b_v = \log v.$$

Define

$$I_{u,v}(H) := \#\{x \in [H, H+1) : u^x, v^x \in \mathbb{Z}\}.$$

Every monomial restricts to the frequency $i \log u + j \log v$, and the exact block scales u^H, v^H are integers.

Theorem 15.1 (Two-base block theorem). *For multiplicatively independent integers $u, v > 1$,*

$$\limsup_{H \rightarrow \infty} I_{u,v}(H) \frac{(\log H)^2}{H^2} \leq \frac{32 \log u \log v}{9}. \quad (48)$$

Proof. Repeat Sections 4–7 with frequencies $i \log u + j \log v$ and the weighted simplex $i \log u + j \log v \leq L$. $\square$

The constant is smallest when the logarithms are unbalanced, but the conjectural structure of counterexamples also changes. The theorem is best viewed as a general sparse-curve counting principle rather than evidence that one pair of bases is intrinsically easier.

16 Proof-producing finite certificates

The asymptotic theorem is useful conceptually, but a finite verifier can extend zero-free regions and provide data about the full successive-minimum spectrum. The essential principle is separation of *discovery* from *verification*: LLL, floating-point Bessel values, and support heuristics may discover a candidate; the final certificate is checked by exact integer arithmetic plus outward-rounded intervals.

16.1 Certificate contents

For each interval $J = [\alpha, \beta] \subset [0, 1]$, store:

1. the integer scale H and rational endpoints α, β ;
2. the ordered monomial support E ;
3. integer coefficient vectors $c^{(1)}, \dots, c^{(r)}$;
4. an exact nonzero $r \times r$ minor certifying their rank;
5. rational intervals enclosing the first N Chebyshev coefficients of every restricted monomial;
6. rational upper bounds for all tails;
7. a rational number $U_a < 1$ satisfying

$$\sum_{n < N} \left| \sum_e c_e^{(a)} a_{n,e} \right| + \sum_e |c_e^{(a)}| \text{Tail}_e \leq U_a.$$

The verifier never needs to reproduce the LLL reduction. It checks the integer minor, the coefficient enclosures, and the displayed rational inequality. Theorem 5.1 then returns the point bound $|E| - r$.

16.2 Directed rounding

The only transcendental constants are $\log 2$, $\log 3$, exponentials, and Bessel values at positive arguments. All admit rapidly convergent monotone series or standard interval-library implementations. A robust verifier can use either:

- MPFR/Arb directed rounding to produce rational dyadic endpoints; or
- a small standalone rational checker for truncated Taylor/Bessel series with a proved remainder.

The second option is slower but minimizes the trusted code base and is well suited to Lean replay.

16.3 Discovery algorithm

A practical search can use the following pipeline.

```

INPUT: scale H, interval J, target defect d_target
1. Choose a weighted-frequency support E and Chebyshev cutoff N >= |E|.
2. Compute high-precision enclosures for the Bessel coefficient matrix.
3. Build an integer-scaled approximation lattice, including tail rows.
4. Run LLL/BKZ to obtain many candidate coefficient vectors.
5. Sort candidates by rigorously verified curve sup bound.
6. Retain vectors with bound < 1 and compute their exact Z-rank.
7. If |E| - rank <= d_target, emit a certificate for J.
8. Otherwise split J or modify E, N, and the lattice scaling.
OUTPUT: exact certificate or a diagnostic successive-minimum profile.

```

For finite exclusion of a nonintegral point in a unit block, one can account separately for the known integral point $(2^H, 3^H)$. A cover defect of at most one proves that there is no additional point. A defect of zero on the open interval $(0, 1)$ is even cleaner.

16.4 Why this can outperform point scanning

A direct scan examines candidate integers one at a time. A trap certificate excludes an entire continuum interval at once. The cost is shifted from the number of possible X values to the dimension of a structured lattice, which grows polynomially in H . The asymptotic theorem does not yet make this polynomial small enough for a proof, but for finite H the constant and the observed rank can be far better than the worst-case estimates.

Computational experiment 16.1 (Certificate reconnaissance). [\[proposed computation\]](#) For each $14 \leq H \leq 40$, compute the full verified rank curve

$$r \longmapsto \mu_r(H)$$

for several weighted-simplex and thin-shell supports. Record:

1. the certified defect $q - r$ at threshold;

2. the distribution of coefficient supports and signs;
3. common factors among short polynomials;
4. the change under $H \mapsto H + 1$;
5. the singular spectrum of the fixed Bessel matrix.

The aim is not merely to enlarge a finite bound, but to decide empirically which of Targets [10.1](#), [14.2](#), [11.4](#), or [12.2](#) has the right shape.

17 Lean formalization plan

The core deterministic theorem is unusually formalization-friendly. It separates into small modules with few analytic dependencies.

17.1 Phase I: exact algebra

1. **ExpFrequency**: prove injectivity of $(i, j) \mapsto i \log 2 + j \log 3$ from unique factorization.
2. **ExpVandermonde**: formalize rank of $(e^{\lambda_j t_i})$ for ordered distinct nodes and frequencies. A Rolle-theorem proof avoids heavy total-positivity libraries.
3. **CurveEvaluationRank**: package Corollary [4.4](#).
4. **RankCodimensionTrap**: prove Theorem [5.1](#) from integer-valuedness and matrix rank.
5. **DilationRigidity**: formalize Theorem [11.1](#) as an ordinary Vandermonde argument over $\mathbb{R}$.

17.2 Phase II: certificate checker

The checker need not formalize LLL or Minkowski. Its input is a finite list of integer vectors and rational interval bounds. Required modules are:

1. rational interval arithmetic;
2. exact determinant/rank checking over $\mathbb{Z}$;
3. a verified bound for Chebyshev polynomials on $[-1, 1]$;
4. a verified exponential/Bessel tail inequality;
5. interval cover bookkeeping.

This phase can certify concrete finite exclusions independently of the asymptotic existence proof.

17.3 Phase III: asymptotic theorem

Formalizing Theorem [7.3](#) requires:

1. Bernstein-ellipse coefficient bounds;
2. Cauchy–Binet and Hadamard estimates;
3. Minkowski’s second theorem for Euclidean lattices;
4. the weighted-simplex lattice-point asymptotics;
5. routine logarithmic asymptotics for the choice (27).

This is substantial but modular. The recommended order is to formalize the rank–codimension checker first, because it can validate computational certificates long before the general Minkowski theorem is available in the repository.

18 Comparison with the inherited methods

Method	Arithmetic object	Best scale exposed	Remaining obstruction
Four exponentials	2×2 exponential matrix	Exact qualitative barrier	Conjectural transcendence theorem
Determinant/ p -adic methods in Report IV	Integer semigroup determinants	Sharp termwise and cancellation ceilings	Missing second source of size
O-minimal counting	Rational points on fixed arc	Polylogarithmic upper bounds	Exponent too large; denominator support ignored
Brisebarre–Hanrot general algorithm	Two small auxiliary polynomials	Efficient local point isolation	Possible common factor
This report: one sparse trap	Exponential Chebyshev sum	$H^2/(\log H)^2$ block bound	Still superlinear
This report: rank–codimension	Full short-vector subspace	Exact defect $q - r$	Need $> 98.59\%$ rank
This report: Bessel/PGN	Successive-minimum spectrum	Exact diagonal-flow factorization	Bulk singular-value theorem missing
Hermite jets	Partial-derivative trap	Potential multiplicity gain	Quadratic jet-row reset tax
Stable factors	Dilation-covariant factor families	Could convert failure into rigidity	Unit-equation bridge not yet built

The new route is complementary to Report IV’s determinant program. Both manufacture small integer-valued analytic objects, but they organize arithmetic differently. The previous program studies determinants of values at many semigroup points; the present program studies many coefficient vectors small on an entire curve segment. Their shared barrier is now visible as a

rank statement: local volume decay is easy, while broad, correlated cancellation across almost all directions is hard.

19 Prioritized research program

The calculations above narrow the next work to five testable programs. They are ordered by expected information gain, not by apparent elegance.

19.1 Priority A: build the deterministic finite verifier

Implement Section 16 first. This has three immediate payoffs:

1. it can extend the finite zero-free region without enumerating every integer candidate;
2. it measures the actual rank at the strict threshold, which the asymptotic covolume cannot predict;
3. it reveals whether common factors, stable supports, or Bessel condition numbers are the real finite-scale phenomenon.

Kill criterion. If optimized supports through the feasible range consistently have rank defect comparable to q , then near-full-rank closure is implausible without new arithmetic rows. If the defect is small or decreases rapidly with H , the PGN program becomes the highest priority.

19.2 Priority B: analyze the fixed Bessel matrix

The factorization (47) isolates the only non-diagonal analytic object. Compute and prove:

- signs and magnitudes of all contiguous minors;
- asymptotic singular-value density for weighted-simplex frequencies;
- variation-diminishing or Neville-elimination factorizations;
- compound-matrix bounds uniform in q .

Kill criterion. An exponentially broad singular-value distribution with a positive fraction above the trap threshold would show that analytic total positivity alone cannot meet the high-corank target.

19.3 Priority C: add arithmetic rows tied to a primitive pair

The unconditional lattice sees only the curve and therefore treats the integral branch and a hypothetical second generator symmetrically. A closing proof must likely add rows that exist only when integer coordinates have the fixed semigroup form $(2^n M^k, 3^n A^k)$. Candidate rows include:

- congruences at structural primes dividing MA but not 6;

- common-factor data from adjacent normalized orbit points;
- exact valuation constraints on coefficients evaluated over a whole nested block;
- transported factors under $(X, Y) \mapsto (2X, 3Y)$.

The objective is not more covolume in a few directions. It is to lift the bottom of the successive-minimum spectrum, so that a determinant bound forces the remaining directions to be short.

Kill criterion. If every added arithmetic row is supported on only $o(q)$ coefficient directions or contributes only $O(H)$ total logarithmic weight, it cannot alter the 98.59% rank requirement.

19.4 Priority D: solve the jet min-plus problem

Target 13.3 is the only route in this report that could change the point-count exponent directly. Form the combined derivative–Chebyshev matrix, assign to a row (α, n) the leading cost

$$H(\alpha_1 \log 2 + \alpha_2 \log 3) + n \log \rho,$$

and compute the minimum total cost of a full-rank row selection. The exact falling-factorial minors decide which assignments are admissible.

Positive outcome. A multiplicity $s \gg H/(\log H)^2$ with a support of size $O(H^2/(\log H)^2)$ proves the conjecture.

Negative outcome. A matching assignment lower bound showing $s = O(H/(\log H)^2)$ closes the most direct Hermite escape and supplies a reusable no-go theorem.

19.5 Priority E: stable factors and unit equations

Run factorization on every certified short polynomial and track factors after dilation normalization. If a bounded-complexity factor family recurs, attempt the Wang–Xiao integral-point degeneration route. If factors do not recur, the data supports a pure PGN model instead.

20 Conclusion

The attached report showed that the conjecture lies exactly on the four-exponentials frontier and closed many plausible determinant, p -adic, cyclotomic, and dynamical shortcuts. The new 2026 Coppersmith–Chebyshev input does not magically cross that frontier, but it changes the shape of the search in four concrete ways.

First, the problem is now an exact rational-grid problem on one fixed analytic arc. Second, strict total positivity removes the generic coprimality heuristic and turns every independent short vector into one unit of point-count codimension. Third, optimizing the frequency support and the Chebyshev ellipse gives a new unconditional logarithmic-square saving:

$$I(H) \leq (2.707555926 \dots + o(1)) \frac{H^2}{(\log H)^2}.$$

Fourth, optimizing over macroscopic windows converts the residual gap into a precise spectral target: at the present kernel frontier, more than 98.59% of the coefficient space must be certified short.

That target is difficult, but it is neither vague nor equivalent to simply restating the conjecture. It can be attacked computationally, through Bessel compound matrices, through parametric geometry of numbers, through structural-prime rows, or through controlled multiscale covariance. Each attack has an exact success criterion and an exact failure mode. The next decisive experiment is therefore clear: build the proof-producing verifier, measure the full successive-minimum spectrum, and let the observed defect determine which theorem to pursue.

A A Rolle-theorem proof of the sparse zero bound

For formalization it is useful to avoid invoking the general theory of total positivity. We record the elementary induction.

Theorem A.1 (Exponential sums have few zeros). *Let $\lambda_1 < \dots < \lambda_q$ and*

$$F(t) = \sum_{j=1}^q c_j e^{\lambda_j t}$$

be nonzero. Then F has at most $q - 1$ real zeros counted with multiplicity.

Proof. Induct on q . The assertion is clear for $q = 1$. Suppose it holds for $q - 1$ and divide by the first exponential:

$$G(t) = e^{-\lambda_1 t} F(t) = c_1 + \sum_{j=2}^q c_j e^{(\lambda_j - \lambda_1)t}.$$

If G has m zeros counted with multiplicity, then G' has at least $m - 1$ zeros counted with multiplicity. This follows by applying Rolle between distinct zeros and reducing each positive multiplicity by one at the same location. But

$$G'(t) = \sum_{j=2}^q c_j (\lambda_j - \lambda_1) e^{(\lambda_j - \lambda_1)t}$$

is either zero or a nontrivial sum of at most $q - 1$ exponentials. If it is nonzero, induction gives $m - 1 \leq q - 2$. If it is zero, then all c_j for $j \geq 2$ vanish and G is a nonzero constant, so $m = 0$. Thus $m \leq q - 1$. $\square$

The same induction proves nonsingularity of every square evaluation matrix: a nonzero kernel vector would give a nontrivial sum with at least as many distinct zeros as terms.

B Detailed fixed-rank successive-minimum estimate

Let B be the matrix (12), let $\Delta = \sqrt{\det(B^T B)}$, and let $\mu_1 \leq \dots \leq \mu_q$ be the Euclidean successive minima. The volume of the unit ball is

$$\omega_q = \frac{\pi^{q/2}}{\Gamma(q/2 + 1)}, \quad -\log \omega_q = \frac{q}{2} \log q + O(q).$$

Minkowski gives

$$\mu_1 \cdots \mu_q \leq \frac{2^q \Delta}{\omega_q}. \tag{49}$$

Because $(0, 0) \in E$, the smallest diagonal tail weight is

$$\tau_{0,0} = \frac{2\rho^{-q}}{1 - \rho^{-1}} \geq 2\rho^{-q}.$$

For every nonzero $c \in \mathbb{Z}^E$, $\|Bc\|_2 \geq 2\rho^{-q}$. Hence, for fixed r ,

$$(2\rho^{-q})^{r-1} \mu_r^{q-r+1} \leq \mu_1 \cdots \mu_q.$$

Together with (49),

$$\begin{aligned} \log \mu_r &\leq \frac{1}{q-r+1} (\log \Delta + (r-1)q \log \rho + O(q \log q)) \\ &\leq \frac{H+R}{q-r+1} \sum_e \lambda_e - \frac{q(q-1)}{2(q-r+1)} \log \rho + \frac{(r-1)q}{q-r+1} \log \rho + O_r(\log q). \end{aligned}$$

For fixed r , replacing $q-r+1$ by q in the leading terms changes the expression by $O_r((H\bar{\lambda})/q + \log \rho)$, which is lower order in the applications. This yields (16).

C Numerical constants and optimization data

All logarithms in this report are natural. The constants are:

$$\begin{aligned} \log 2 &= 0.6931471805599453094\dots, \\ \log 3 &= 1.0986122886681096914\dots, \\ \frac{32 \log 2 \log 3}{9} &= 2.707555592600207\dots \end{aligned}$$

The root of $t \tanh t = 2$ is

$$t_0 = 2.065338138974704728\dots, \quad \rho_0 = e^{t_0} = 7.887964675046528\dots$$

Then

$$\begin{aligned} R(\rho_0) &= 2.503685361986428\dots, \\ r_0 &= \frac{1}{R(\rho_0) - 1} = 0.665032892479158\dots, \\ \kappa_* &= 5.087266580727214\dots \end{aligned}$$

The rank thresholds are summarized below.

Input on least generator	$\kappa_* \beta$	Required rank fraction	Maximum defect fraction
Kernel level $\beta \geq 14$	71.221732...	0.98595934...	0.01404066...
Software level $\beta > 27$	137.356198...	0.99271966...	0.00728034...

For comparison, using the unweighted total-degree triangle would give the larger window constant

$$\frac{16(\log 6)^2}{9} \frac{\cosh t_0}{t_0^2} = 5.361841943267\dots$$

The weighted simplex reduces it by the exact factor $4 \log 2 \log 3 / (\log 6)^2 = 0.948\dots$

D A machine-checkable certificate schema

A compact textual schema can be organized as follows.

```
certificate_version: 1
scale_H: integer
intervals:
  - left: rational
    right: rational
    support: [[i_1,j_1],...,[i_q,j_q]]
    chebyshev_cutoff: N
    polynomials:
      - coefficients: [c_1,...,c_q]
        rank_minor_row: optional
        verified_upper_bound: rational < 1
    rank_minor:
      row_indices: [...]
      column_indices: [...]
      determinant: nonzero integer
    coefficient_intervals:
      # interval enclosure for each n,e
    tail_bounds:
      # nonnegative rational for each e
claimed_point_bound: sum(q-r)
```

A checker performs no search. It verifies:

1. every support exponent is nonnegative;
2. coefficient vectors and the stored minor agree;
3. the minor determinant is nonzero;
4. interval arithmetic proves each displayed upper bound;
5. intervals cover the requested parameter range;
6. the claimed point bound equals the sum of local defects.

The mathematical kernel is Theorem 5.1; all other operations are finite.

E Dependency graph of the new results

Result	Dependencies
Sparse zero bound	Multiplicative independence of 2, 3; Rolle’s theorem
Rank–codimension trap	Sparse evaluation rank; integer-valuedness
Norm-to-sup transfer	Chebyshev expansion; Cauchy–Schwarz
Covolume bound	Bernstein coefficient bound; Cauchy–Binet; Hadamard
Fixed-rank trap	Covolume bound; Minkowski’s second theorem
Weighted block theorem	Fixed-rank trap; weighted simplex count and first moment
Window constant	Weighted threshold; elementary two-variable optimization
Dilation rigidity	Ordinary Vandermonde determinant
Hermite trap	Integer derivatives; sparse zero multiplicity
Finite certificate checker	Rank–codimension theorem; rational interval arithmetic

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